Desk Companion

Plan your day by stacking it on your real desk, and see overcommitment before you live it.

Productivity track | New Experience | Target: Best New Productivity Experience

AT A GLANCE

Track / division	Productivity (habit tied to morning routine and wind-down) / New Experience
Primary prize	Best New Productivity Experience ($70,000; runner-up $35,000)
Fallback special award	Best Agentic Interaction ($25,000) or Best Reason to Come Back ($25,000)
Session length	Morning ritual about 5 minutes; wind-down about 3 minutes
Room requirement	A desk or table; tap-to-place fallback on any flat surface
Core tech	Plane detection and hit-test placement, hand-pinch grabbing, IndexedDB, Claude via a Cloud Run endpoint
Biggest risk	'Why put on a headset to plan my day?'
Signature moment	The stack rising past the red capacity line
Status (2 Oct 2026)	Full loop built and live; awaiting first physical Quest test
Live build	desk-companion-1031148889398.us-central1.run.app
Repository	github.com/4KInc/desk-companion (pull request #1)

The one-line pitch

A mixed-reality daily planner where tasks have physical size. Each task becomes a tile whose thickness is its time estimate, and you sort the tiles by hand into zones on your real desk. Stack more than your day can hold and the pile rises past a glowing capacity line: overcommitment you can see and feel.

SECTION 01

Status report: 2 October 2026

Where we are in one paragraph

The full Desk Companion loop is built, verified in the IWER emulator from real input, and live on a public HTTPS URL. Every must-ship item from the original scope is done, and most of the nice-to-haves too: focus mode, estimation-bias learning, weekly jar strata, a guided first run and sound design. The build is roughly five weeks ahead of the original schedule with 47 days to the deadline. The one large unknown is the physical headset: nothing has run on a Quest yet, so hand-pinch feel, the flick threshold, dictation support and real frame rate are unverified.

Live build	https://desk-companion-1031148889398.us-central1.run.app (Cloud Run, us-central1, revision 2)
Code	github.com/4KInc/desk-companion, branch feat/desk-companion-core, pull request #1 into main (open)
Agent	Claude Opus 5.5 (low effort, structured output) via a server-side endpoint; key in Google Secret Manager
Deadline	18 Nov 2026, 12:00 PT. Plan: submit 17 Nov. 47 days remaining
Biggest open risk	No physical Quest test yet

What is implemented

Area	Feature	Status	How it was verified
Placement	Tray placed on the Space Setup 'table' plane in front of the user, facing them	Done	Emulator, living room and music room
Placement	Pinch-to-place fallback and a Move tray button	Done	Emulator: reticle followed the ray, pinch placed the tray
Core loop	Tiles 1 cm thick per 30 min; Now / Later / Drop zones with snapping and auto-stacking	Done	Emulator, real input
Core loop	Grab by pinch; only the top tile of each pile is grabbable	Done	Emulated controllers; hand pinch untested on device
Core loop	Capacity line; overflow (judged in minutes) turns line and top tiles red and wobbling	Done	Emulator, real input
Core loop	Done jar that never resets	Done	Emulator
Agent	Brain dump: 2D text box or in-XR dictation, split by Claude into titled, estimated, prioritised, categorised tasks	Done	Real Claude calls in dev and production
Agent	Offline splitter when the agent is unreachable	Done	Emulator and logic check

Area	Feature	Status	How it was verified
Agent	Capacity nudge: smallest set of lowest-priority tiles to move to Later, Move / Keep	Done	Emulator, both buttons
Agent	Split a big task: a 2h+ tile landing in Now gets an offer; Claude cracks it into 2 to 4 steps in place	Done	Emulator, real Claude call
Safety	Undo (8 s) after a flick to tomorrow, a jar drop or a move into Drop; flicks need speed and 12 cm of travel	Done	Emulator: flick and jar both undone
Grab feel	Hover lift: the tile nearest a hand rises 1.5 cm and glows, so the pinch closes in free air	Done	Emulator: measured 1.5 cm lift
Learning	Focus mode with a floating timer; plausible focus time becomes the actual	Done	Emulator, real seconds counted
Learning	'How long did it take?' card; per-category bias; recap line; adjusted new estimates	Done	Emulator plus logic tests
Daily rhythm	Flick left to push to tomorrow; Wind down recap; End day, history and streak; rollover	Done	Emulator plus logic tests
Retention	Weekly strata in the done jar (current week glowing on top)	Done	Logic tests; one layer visible live
Onboarding	Guided first run: ghost tile, pulsing rings, one-line coach hints	Done	Emulator, all five steps
Polish	Procedural spatial sound design (no audio files)	Done	Event trace; not listened to on device
Hosting	Cloud Run server, Secret Manager key, request cap and per-IP rate limit	Live	Anonymous requests to the public URL
Performance	Two draw groups per tile, room meshes off	Done	102 draw calls with 13 tiles (was 206)
Persistence	Tray position remembered across sessions with anchors	Not built	Tray re-detects the desk each session
Store	PWA manifest, icons, Quest APK packaging	Not started	Optional; the submission only needs a URL
Accessibility	Options card: larger tiles, high contrast, reduce motion, sound; button fallbacks for every gesture	Done	Emulator, hand input; settings persist
Polish	Rounded tiles (two draw groups kept)	Done	Emulator screenshot

Verification so far

- **Logic tests:** four scripts covering day rollover and streaks (including month boundaries), weekly jar layers, the estimation-bias model and the coach steps. All pass.
- **Live emulator runs:** every feature above was exercised through real XR input (ray clicks, squeeze-grab, flick swings, chest release) with named ECS and UI state checked after each step.
- **Production:** page, assets, MIME types, caching, 404 for secrets, 413 for oversized requests, 429 after 20 requests in 10 minutes, and live Claude calls from the public URL.
- **Performance (desktop emulator):** 102 draw calls and 23k triangles with 13 tiles; median frame 8.3 ms with no dropped frames; the app's own systems cost about 0.09 ms per frame.
- **Not yet verified:** anything on a physical Quest.

What changed from the original plan

Original plan	What was built	Why
Host on Vercel	Google Cloud Run with Secret Manager	The developer chose GCP; one container serves the app and the API.
A fast, cheap model	Claude Opus 5.5 at low effort with structured output	Best quality at acceptable latency (2 to 5 s per brain dump).
Free-form stacking	Snap zones, auto-stacking, top-of-pile grabbing	Thin tiles millimetres apart made proximity grab pick the wrong tile.
Agent writes the nudge	Deterministic nudge from priorities and minutes	Fast, offline-safe and never claims knowledge it lacks; Claude supplies priorities and categories.
Focus mode and bias learning as nice-to-haves	Both built	The schedule ran far ahead; they power the Reason to Come Back story.
Anchors keep the tray on the desk	Desk re-detected each session, plus Move tray	Persistent anchors in Quest Browser are unverified; re-detection takes about 2 seconds.
Room mesh detection	Turned off	Unused by the app; it doubled draw calls.

What is next

Dates	Focus	Done when
2 - 8 Oct	**Quest smoke test** on the live URL: hand pinch on tiles, flick threshold, chest-release radius, label legibility, dictation support, real frame rate with 15+ tiles	Every interaction works by hand on device; issues logged and fixed
9 - 15 Oct	Device fixes; persistent anchors if Quest Browser supports them; large-tile and high-contrast options	Tray reappears on the desk without re-placing (or documented fallback)
16 - 22 Oct	Use it daily for a week (real bias data); art pass on tiles and jar; hover and grab highlights for hands; sound mix on device	A week of real use with no blocking bugs
23 Oct - 5 Nov	Playtests with 5+ people in their own rooms; first-five-minutes iteration; **start recording video footage 26 Oct**	Cold testers finish the morning ritual unaided

Dates	Focus	Done when
6 - 12 Nov	Edit the under-3-minute video; final submission text; optional PWA manifest and icons	Video and text ready
13 - 17 Nov	Freeze, final deploy, verify the public URL, submit on 17 Nov	Submitted a day early

Open decisions and housekeeping

- **Endpoint hardened (3 Oct):** rate limit keyed on the real client IP (last X-Forwarded-For entry), same-origin check on POSTs, a global daily cap, and no task text in logs.
- **Rotate the Anthropic API key:** the current key was pasted into a chat. Create a new one, update .env and the Secret Manager secret, then redeploy.
- **Merge pull request #1** into main once reviewed.
- **gcloud account:** the developer's owner account is currently active; switch back to the service account when appropriate.
- **Cost:** Cloud Run scales to zero with at most 2 instances; the brain-dump endpoint is limited to 20 requests per visitor per 10 minutes.
- **Store packaging:** decide whether to ship a Horizon Store PWA in addition to the URL.

SECTION 02

The idea

To-do apps are lists, and lists hide the thing that actually hurts: you planned nine hours of work into a five-hour day. In mixed reality, tasks can have physical size. Put too much on your desk and you see the pile overflow.

Desk Companion is not a to-do list in VR. It is your workload made physical, sitting on your actual desk. It is designed around two short, recurring moments: a 5-minute morning plan and a 3-minute evening wind-down.

The core mechanic

Every task is a tile with real thickness: a 30-minute task is a thin tile, a 3-hour task a thick block (for example 1 cm per 30 minutes, so an 8-hour day is a 16 cm stack). A translucent capacity line at the Now zone is set from the hours you actually have today. Stack past it and the top tiles glow red and wobble. You feel the overcommitment instead of reading about it.

SECTION 03

How a day works

Morning ritual (about 5 minutes)

1. **The tray appears.** Put on the headset at your desk. The app finds the table surface and places a soft glowing day tray on it, anchored to your real desk.
2. **Brain dump.** Dictate or type what is on your mind. The agent splits it into task tiles, each with a time estimate. Tiles drop onto the desk.
3. **Sort by hand.** Pinch-grab tiles into three zones drawn on the desk: Now, Later and Drop. Tiles snap into place and stack automatically.
4. **Capacity line.** As the Now stack grows it approaches the line set by today's available hours. Past it, the top tiles glow red and wobble.
5. **Agent nudge.** A short suggestion appears beside the stack, such as 'Now is 1.5 h over. These two are lowest priority. Move them to Later?' One pinch accepts.

During the day (optional)

Pull a tile toward your chest and it becomes a focus card, with a timer floating above the desk while the other tiles dim.

Wind-down (about 3 minutes)

- Sweep finished tiles into a done jar on the corner of your desk. The jar fills across days and weeks and never resets.
- For unfinished tiles, flick left to push them to tomorrow, or drop them in Drop.
- For each finished tile, a quick slider asks roughly how long it took (optional, one pinch).
- The agent gives a one-line recap: 'You planned 6 h and did 4.5 h. Your estimates on writing tasks run about 40% low.'

SECTION 04

Improvements over the first draft of this concept

Original plan	Change	Why
Free-form stacking by hand	Tiles snap into zones and stack automatically; the pinch only chooses the zone.	Precise stacking with hand tracking is fiddly and a failed drop ruins the demo.
Agent notices dependencies like 'waiting on a colleague's review'	Nudges only use information the user actually gave: estimates, priority, capacity, and explicitly stated dependencies.	Judges will test the agent; it should never claim knowledge it cannot have.
Estimation-bias learning as a headline feature	Show it in the video as a feature that grows over weeks; do not headline it.	It needs weeks of real data that will not exist by the deadline.

Original plan	Change	Why
Voice input assumed	Voice if available; otherwise a companion phone page (scan a code) or quick-pick templates.	Speech recognition inside a WebXR session in Quest Browser is unverified.
Plane detection assumed	Tap-to-place on any flat surface is a first-class path.	A table is only found if the user labelled it during Space Setup.

SECTION 05

Why passthrough and the real room matter

- **The content lives on your desk.** The tray, zones and stacks sit on your real work surface, at real scale. With passthrough off the concept breaks, which is what the judges want.
- **Physical size carries meaning.** Height on your desk is a real measurement you can compare with the objects around you: 'today's plan is taller than my coffee mug' is an intuitive signal.
- **The done jar is a permanent object in your room.** It sits on your desk corner every day, filling up.

SECTION 06

The first five minutes

Time	What happens
0:00	Passthrough. A soft light traces the edge of the desk and a day tray settles onto it.
0:20	A prompt: 'What is on your mind today?' The user speaks or uses the fallback input.
1:00	Tiles drop onto the tray one by one, each labelled with a title and time estimate.
1:30	Three zones glow: Now, Later, Drop. The first tile pulses; the user pinches it and drops it in Now. It snaps and stacks.
3:00	The Now stack approaches the capacity line; the line brightens. One more tile and it glows red and wobbles.
3:30	The agent nudge appears; one pinch moves two tiles to Later and the stack settles under the line with a satisfying sound.
4:30	Done. 'See you this evening.' The first run is the tutorial, done with the user's own tasks.

SECTION 07

The agent

The AI makes the experience faster, but the experience stands without it, which satisfies the take-it-away test: tiles can be created manually, sorted physically, checked against capacity, and swept into the jar with no AI at all.

Agent job	What it does	Visible as	Status
Split the brain dump	Claude turns free text into tasks with a title, estimate, priority and category	Tiles dropping into the Inbox	Built
Capacity nudge	Picks the smallest set of lowest-priority tiles that fixes an overflow	A card on the panel with Move / Keep	Built (deterministic, uses Claude's priorities)
Evening recap	Planned versus finished, tomorrow pile, dropped, carried over, bias line	The Wind down card	Built (local template)
Estimation bias	Per-category actual/estimate ratio from the last 20 tasks	Recap line and adjusted new estimates	Built
Split a big task	Offered when a 2h+ tile lands in Now; Claude names 2 to 4 concrete steps	The tile cracks and its steps fly into the pile	Built (offline: equal parts)

Implementation

- A server-side endpoint (POST /api/brain-dump on the Cloud Run service, shared with the Vite dev server) holds the API key so it never reaches the headset.
- It calls Claude Opus 5.5 at low effort with structured output (a fixed schema), with server-side refusal fallbacks, so the app always receives valid task data. Typical latency is 2 to 5 seconds.
- The app sends only the text needed for the request and stores all tasks locally in IndexedDB. No accounts and no server-side storage.
- If the network or AI is unavailable, the app falls back to manual tile creation with preset durations (15 min, 30 min, 1 h, 2 h).

SECTION 08

Progression and reasons to come back

System	Description
The done jar	Fills with finished tiles across days and weeks and never resets. The main visible growth.
Streaks	Consecutive days with both a morning plan and a wind-down.
Honest estimates	Over weeks, the app learns your estimation bias per category and shows how your planning improves.
Weekly shelf	At the end of each week, the jar's contents compress into a labelled layer, so months of work are visible as strata.

SECTION 09

Accessibility

- Voice input where available, with typing and preset alternatives.
- Large tiles and generous snap zones for limited fine motor control.
- Capacity warnings use motion, height and pattern, not only red colour.
- Captions for all audio; high-contrast mode for tile text.

SECTION 10

Art direction and sound

Soft, tactile and calm: frosted-glass tiles with clear typography, gentle glows for zones, a warm glass jar. The overflow moment is the one dramatic beat: red glow, wobble and a low hum. Sounds are soft clicks for snapping, a glassy chime for tiles entering the jar.

SECTION 11

Technical architecture (IWSDK)

Feature	Used for
Plane detection	Finding the desk (the scaffold already enables it).
Hit test	Tap any surface to place the tray when no table is found.
Anchors	Keeping the tray and jar in place across days (persistence in Quest Browser needs checking).
Grabbing and hand tracking	Picking up and dropping tiles; flick and sweep gestures.
UI (UIKit)	Tile text, nudge cards, recap text.
IndexedDB	All task data, history and jar contents.

Feature	Used for
Serverless function	The agent calls.

System	Responsibility
DeskSetupSystem	Places the tray and jar; saves the anchors.
TaskTileSystem	Creates tiles from task data; sets thickness from the estimate.
ZoneSnapSystem	Snaps dropped tiles into zones and restacks them.
CapacitySystem	Compares stack height with the capacity line; triggers the overflow effects.
AgentSystem	Calls the serverless function and turns responses into tiles and nudges.
FocusSystem	The focus card and timer.
JarSystem	Done jar, streaks and weekly strata.

SECTION 12

Risks and mitigations

Risk	Severity	Mitigation
'Why a headset to plan my day?'	High	Target people who already work in a headset; make the overflow moment striking in the first 20 seconds of the video.
Productivity is a crowded idea	High	Lead with the capacity line and the done jar, not feature breadth.
Voice input unavailable in WebXR	Medium	Phone companion page or presets; test in week 1.
Fiddly grabbing	Medium	Snap zones and automatic stacking.
Anchor persistence unreliable	Medium	Re-detect the desk each session; store jar contents in IndexedDB regardless.
Agent overreaches	Medium	Structured output; nudges based only on known data.

Week-1 checks: status

Question	Answer so far
Does the Web Speech API work inside a WebXR session in Quest Browser?	Works in the desktop emulator; unverified on Quest. The Dictate button hides itself when unsupported, and the 2D brain-dump box is the fallback.
Do persistent anchors survive closing and reopening the app?	Not tested; not used. The tray re-detects the desk each session, and Move tray re-places it by hand.
Does plane detection return a labelled table in your own room?	Yes in two emulated rooms; real rooms unverified. Pinch-to-place covers rooms without a labelled table.

Question	Answer so far
Can hand tracking reliably pinch a 6 cm tile lying on a desk?	Unverified on device. Tiles are 10 x 7 cm, pinch-to-grab is enabled, and only pile tops are grabbable.

SECTION 13

Original seven-week build plan, with progress

The original plan is kept for reference. Content planned for weeks 2 to 5 was finished in the first week, in the emulator; the remaining weeks shift to device testing, playtesting and the video (see 'What is next' above).

Week	Original goal	Status on 2 Oct
1	Platform checks (speech, anchors, planes, pinch); tray placement	Placement done; device checks pending
2	Tiles with thickness; zones; snapping and stacking	Done
3	Capacity line and overflow; agent function for brain dump	Done
4	Agent nudge; first-run flow; IndexedDB persistence	Done
5	Wind-down: done jar, flick to tomorrow, recap; focus card	Done
6	Performance; playtests over several days; accessibility options	Performance pass done (desktop); playtests and accessibility next
7	Polish; record video; write submission	Not started
Ship	Final checks; submit on 17 Nov	Planned

SECTION 14

Scope

Must ship	Nice to have	Cut
Desk detection and tray; tiles; three zones; capacity line; done jar; agent brain dump. All built.	Focus mode; estimation-bias learning; streak; weekly strata; sound design. **All built.** Persistent anchors and accessibility options still open.	Calendar integrations; accounts; sync across devices. Still cut.

SECTION 15

Demo video plan (under 3 minutes)

Time	Beat	Shot
0:00 - 0:15	Hook	Tiles piling past the red line on a real desk, wobbling.
0:15 - 1:30	Morning ritual	Real-time brain dump, tiles dropping, sorting by hand.
1:30 - 2:00	The nudge	The agent's suggestion; one pinch; the stack settles below the line.
2:00 - 2:40	Wind-down	Sweeping tiles into the jar; flicking unfinished ones to tomorrow.
2:40 - 2:55	Coming back	Montage of the jar filling across days.

SECTION 16

Submission draft

Name	Desk Companion
Tagline (under 140 characters)	Plan your day by stacking it on your real desk. See overcommitment before you live it.
Track / division	Productivity / New Experience
Target launch	Web release Q1 2027

Description (first draft)

To-do lists hide the problem that hurts most: planning more than a day can hold. Desk Companion gives every task physical size on your real desk.

Each morning, tell Desk Companion what is on your mind. An AI agent turns it into task tiles whose thickness is their time estimate. Sort them by hand into Now, Later and Drop. When the Now stack rises past your capacity line, the top tiles glow red and wobble, and the agent suggests what to move. Each evening, sweep finished tiles into a done jar on your desk that fills up over weeks.

We built it with the Immersive Web SDK: plane detection finds your desk (with a pinch-to-place fallback for any surface), hand-pinch grabbing with a hover lift picks up tiles, and Claude runs behind a small server-side endpoint so the key never reaches the headset. The agent splits your brain dump into estimated tasks, offers to crack big tasks into smaller steps, and learns how long your work really takes. Everything also works without it: you can create, sort and finish tasks entirely by hand.

SECTION 17

How it maps to the judging criteria

Criterion (25% each)	Evidence in Desk Companion
Innovation and Creativity	Workload as physical height; overcommitment made visible and tangible.
Experience Design	Two recurring rituals; first run is the tutorial; content sits low and central.
Technical Implementation	Plane detection with fallback, anchors, grabbing, an AI agent with structured output.
Polish and Presentation	The overflow shot is a strong hook; calm, tactile visuals.

Hard requirement and design guidelines

Test	What the rules ask	How this idea passes
Hands-first (hard requirement)	Complete the entire experience without ever pairing a controller.	Every action is a pinch, grab, sweep or flick.
Airplane seat test	Every interaction works within a two-foot radius, seated.	Everything happens on the desk within arm's reach.
One bus stop test	Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less.	Morning is 5 minutes, wind-down 3 minutes; each is complete on its own.
Take it away test	Remove any third-party service and a project still stands.	Remove the AI and you can still create tiles, sort them, see capacity and fill the jar.

Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.
- **Wrapper rule:** the AI is an accelerator, not the product. Make sure the video shows manual tile creation too.

SECTION 18

Honest weaknesses

- The biggest risk is the question 'why put on a headset to plan my day?'
- Productivity apps are a common idea; others will build task managers too.
- Voice input in Quest Browser needs checking.
- No multiplayer, so the Social award is out of reach (which is fine).

SECTION 19

Scorecard and verdict

Dimension	Score / 10	Comment
Originality	7	The capacity line is original; the category is not.
Only possible in MR	8	Physical size on a real desk is the point.
Works in untested rooms	8	One surface plus a fallback.
Solo feasibility in 7 weeks	7	Moderate; the agent adds an external dependency.
Prize ceiling	$70k	Productivity track; Agentic Interaction as a fallback.

Verdict

A solid, credible Productivity entry built around one strong mechanic. Choose it if you prefer the Productivity track or want to compete for the Agentic Interaction award; focus everything on the overflow moment and the done jar.